

4.7.1 Broadcast Routing Algorithms

Perhaps the most straightforward way to accomplish broadcast communication is for the sending node to send a separate copy of the packet to each destination, as shown in Figure 4.43(a). Given N destination nodes, the source node simply makes N copies of the packet, addresses each copy to a different destination, and then transmits the N copies to the N destinations using unicast routing. This **N-way-unicast** approach to broadcasting is simple—no new network-layer routing protocol, packet-duplication, or forwarding functionality is needed. There are, however, several drawbacks to this approach. The first drawback is its inefficiency. If the source node is connected to the rest of the network via a single link, then N separate copies of the (same) packet will traverse this single link. It would clearly be more efficient to send only a single copy of a packet over this first hop and then have the node at the other end of the first hop make and forward any additional needed copies. That is, it would be more efficient for the network nodes themselves (rather than just the source node) to create duplicate copies of a packet. For example, in Figure 4.43(b), only a single copy of a packet traverses the R1-R2 link. That packet is then duplicated at R2, with a single copy being sent over links R2-R3 and R2-R4.

The additional drawbacks of N -way-unicast are perhaps more subtle, but no less important. An implicit assumption of N -way-unicast is that broadcast recipients, and their addresses, are known to the sender. But how is this information obtained? Most likely, additional protocol mechanisms (such as a broadcast membership or destination-registration protocol) would be required. This would add more overhead and, importantly, additional complexity to a protocol that had initially seemed quite simple. A final drawback of N -way-unicast relates to the purposes for which broadcast is to be used. In Section 4.5, we learned that link-state routing protocols use broadcast to disseminate the link-state information that is used to compute unicast routes. Clearly, in situations where broadcast is used to create and update unicast routes, it would be unwise (at best!) to rely on the unicast routing infrastructure to achieve broadcast.

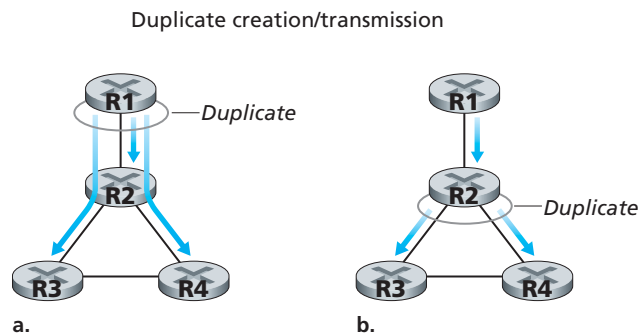


Figure 4.43 ♦ Source-duplication versus in-network duplication

Given the several drawbacks of N -way-unicast broadcast, approaches in which the network nodes themselves play an active role in packet duplication, packet forwarding, and computation of the broadcast routes are clearly of interest. We'll examine several such approaches below and again adopt the graph notation introduced in Section 4.5. We again model the network as a graph, $G = (N, E)$, where N is a set of nodes and a collection E of edges, where each edge is a pair of nodes from N . We'll be a bit sloppy with our notation and use N to refer to both the set of nodes, as well as the cardinality ($|N|$) or size of that set when there is no confusion.

Uncontrolled Flooding

The most obvious technique for achieving broadcast is a **flooding** approach in which the source node sends a copy of the packet to all of its neighbors. When a node receives a broadcast packet, it duplicates the packet and forwards it to all of its neighbors (except the neighbor from which it received the packet). Clearly, if the graph is connected, this scheme will eventually deliver a copy of the broadcast packet to all nodes in the graph. Although this scheme is simple and elegant, it has a fatal flaw (before you read on, see if you can figure out this fatal flaw): If the graph has cycles, then one or more copies of each broadcast packet will cycle indefinitely. For example, in Figure 4.43, R2 will flood to R3, R3 will flood to R4, R4 will flood to R2, and R2 will flood (again!) to R3, and so on. This simple scenario results in the endless cycling of two broadcast packets, one clockwise, and one counterclockwise. But there can be an even more calamitous fatal flaw: When a node is connected to more than two other nodes, it will create and forward multiple copies of the broadcast packet, each of which will create multiple copies of itself (at other nodes with more than two neighbors), and so on. This **broadcast storm**, resulting from the endless multiplication of broadcast packets, would eventually result in so many broadcast packets being created that the network would be rendered useless. (See the homework questions at the end of the chapter for a problem analyzing the rate at which such a broadcast storm grows.)

Controlled Flooding

The key to avoiding a broadcast storm is for a node to judiciously choose when to flood a packet and (e.g., if it has already received and flooded an earlier copy of a packet) when not to flood a packet. In practice, this can be done in one of several ways.

In **sequence-number-controlled flooding**, a source node puts its address (or other unique identifier) as well as a **broadcast sequence number** into a broadcast packet, then sends the packet to all of its neighbors. Each node maintains a list of the source address and sequence number of each broadcast packet it has already received, duplicated, and forwarded. When a node receives a broadcast packet, it first checks whether the packet is in this list. If so, the packet is dropped; if not, the

packet is duplicated and forwarded to all the node's neighbors (except the node from which the packet has just been received). The Gnutella protocol, discussed in Chapter 2, uses sequence-number-controlled flooding to broadcast queries in its overlay network. (In Gnutella, message duplication and forwarding is performed at the application layer rather than at the network layer.)

A second approach to controlled flooding is known as **reverse path forwarding (RPF)** [Dalal 1978], also sometimes referred to as reverse path broadcast (RPB). The idea behind RPF is simple, yet elegant. When a router receives a broadcast packet with a given source address, it transmits the packet on all of its outgoing links (except the one on which it was received) only if the packet arrived on the link that is on its own shortest unicast path back to the source. Otherwise, the router simply discards the incoming packet without forwarding it on any of its outgoing links. Such a packet can be dropped because the router knows it either will receive or has already received a copy of this packet on the link that is on its own shortest path back to the sender. (You might want to convince yourself that this will, in fact, happen and that looping and broadcast storms will not occur.) Note that RPF does not use unicast routing to actually deliver a packet to a destination, nor does it require that a router know the complete shortest path from itself to the source. RPF need only know the next neighbor on its unicast shortest path to the sender; it uses this neighbor's identity only to determine whether or not to flood a received broadcast packet.

Figure 4.44 illustrates RPF. Suppose that the links drawn with thick lines represent the least-cost paths from the receivers to the source (A). Node A initially broadcasts a source-A packet to nodes C and B. Node B will forward the source-A packet it has received from A (since A is on its least-cost path to A) to both C and D. B will ignore (drop, without forwarding) any source-A packets it receives from any other

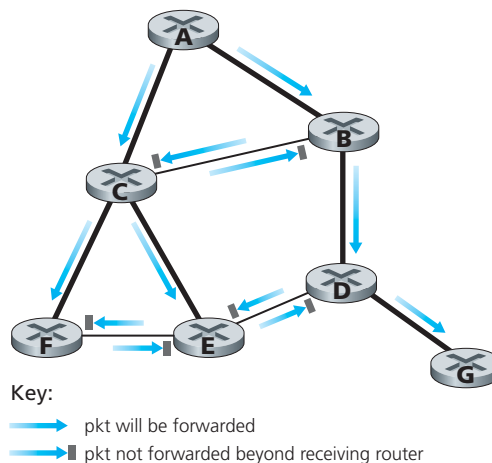


Figure 4.44 ♦ Reverse path forwarding

nodes (for example, from routers *C* or *D*). Let us now consider node *C*, which will receive a source-*A* packet directly from *A* as well as from *B*. Since *B* is not on *C*'s own shortest path back to *A*, *C* will ignore any source-*A* packets it receives from *B*. On the other hand, when *C* receives a source-*A* packet directly from *A*, it will forward the packet to nodes *B*, *E*, and *F*.

Spanning-Tree Broadcast

While sequence-number-controlled flooding and RPF avoid broadcast storms, they do not completely avoid the transmission of redundant broadcast packets. For example, in Figure 4.44, nodes *B*, *C*, *D*, *E*, and *F* receive either one or two redundant packets. Ideally, every node should receive only one copy of the broadcast packet. Examining the tree consisting of the nodes connected by thick lines in Figure 4.45(a), you can see that if broadcast packets were forwarded only along links within this tree, each and every network node would receive exactly one copy of the broadcast packet—exactly the solution we were looking for! This tree is an example of a **spanning tree**—a tree that contains each and every node in a graph. More formally, a spanning tree of a graph $G = (N, E)$ is a graph $G' = (N, E')$ such that E' is a subset of E , G' is connected, G' contains no cycles, and G' contains all the original nodes in G . If each link has an associated cost and the cost of a tree is the sum of the link costs, then a spanning tree whose cost is the minimum of all of the graph's spanning trees is called (not surprisingly) a **minimum spanning tree**.

Thus, another approach to providing broadcast is for the network nodes to first construct a spanning tree. When a source node wants to send a broadcast packet, it sends the packet out on all of the incident links that belong to the spanning tree. A node receiving a broadcast packet then forwards the packet to all its neighbors in the

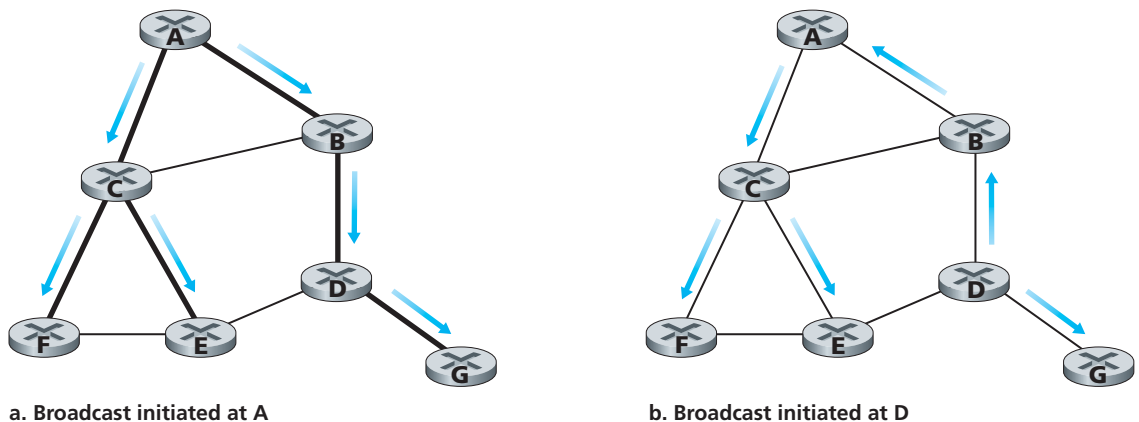


Figure 4.45 ♦ Broadcast along a spanning tree

spanning tree (except the neighbor from which it received the packet). Not only does spanning tree eliminate redundant broadcast packets, but once in place, the spanning tree can be used by any node to begin a broadcast, as shown in Figures 4.45(a) and 4.45(b). Note that a node need not be aware of the entire tree; it simply needs to know which of its neighbors in G are spanning-tree neighbors.

The main complexity associated with the spanning-tree approach is the creation and maintenance of the spanning tree. Numerous distributed spanning-tree algorithms have been developed [Gallager 1983, Gartner 2003]. We consider only one simple algorithm here. In the **center-based approach** to building a spanning tree, a center node (also known as a **rendezvous point** or a **core**) is defined. Nodes then unicast tree-join messages addressed to the center node. A tree-join message is forwarded using unicast routing toward the center until it either arrives at a node that already belongs to the spanning tree or arrives at the center. In either case, the path that the tree-join message has followed defines the branch of the spanning tree between the edge node that initiated the tree-join message and the center. One can think of this new path as being grafted onto the existing spanning tree.

Figure 4.46 illustrates the construction of a center-based spanning tree. Suppose that node E is selected as the center of the tree. Suppose that node F first joins the tree and forwards a tree-join message to E . The single link EF becomes the initial spanning tree. Node B then joins the spanning tree by sending its tree-join message to E . Suppose that the unicast path route to E from B is via D . In this case, the tree-join message results in the path BDE being grafted onto the spanning tree. Node A next joins the spanning group by forwarding its tree-join message towards E . If A 's unicast path to E is through B , then since B has already joined the spanning tree, the arrival of A 's tree-join message at B will result in the AB link being immediately grafted onto the spanning tree. Node C joins the spanning tree next by forwarding its tree-join message directly to E . Finally, because the unicast routing from G to E

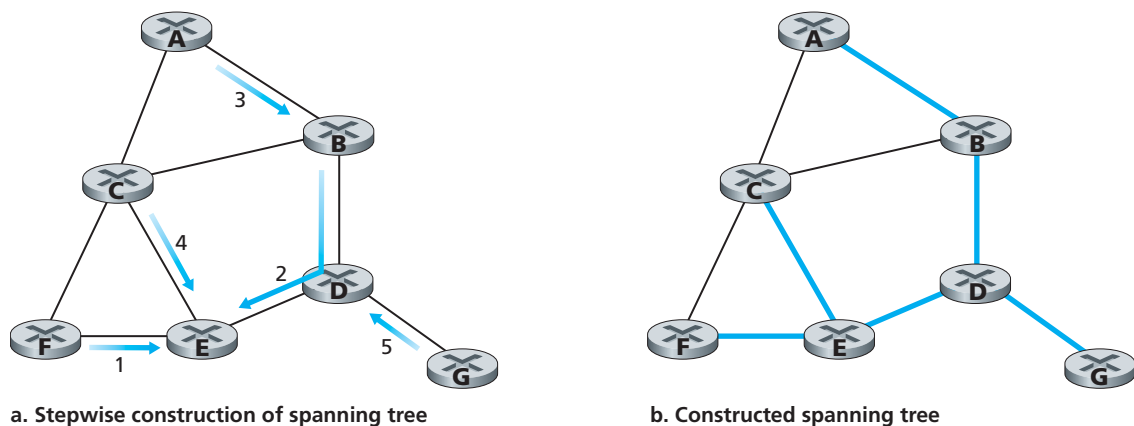


Figure 4.46 ♦ Center-based construction of a spanning tree

must be via node D , when G sends its tree-join message to E , the GD link is grafted onto the spanning tree at node D .

Broadcast Algorithms in Practice

Broadcast protocols are used in practice at both the application and network layers. Gnutella [Gnutella 2009] uses application-level broadcast in order to broadcast queries for content among Gnutella peers. Here, a link between two distributed application-level peer processes in the Gnutella network is actually a TCP connection. Gnutella uses a form of sequence-number-controlled flooding in which a 16-bit identifier and a 16-bit payload descriptor (which identifies the Gnutella message type) are used to detect whether a received broadcast query has been previously received, duplicated, and forwarded. Gnutella also uses a time-to-live (TTL) field to limit the number of hops over which a flooded query will be forwarded. When a Gnutella process receives and duplicates a query, it decrements the TTL field before forwarding the query. Thus, a flooded Gnutella query will only reach peers that are within a given number (the initial value of TTL) of application-level hops from the query initiator. Gnutella's flooding mechanism is thus sometimes referred to as *limited-scope flooding*.

A form of sequence-number-controlled flooding is also used to broadcast link-state advertisements (LSAs) in the OSPF [RFC 2328, Perlman 1999] routing algorithm, and in the Intermediate-System-to-Intermediate-System (IS-IS) routing algorithm [RFC 1142, Perlman 1999]. OSPF uses a 32-bit sequence number, as well as a 16-bit age field to identify LSAs. Recall that an OSPF node broadcasts LSAs for its attached links periodically, when a link cost to a neighbor changes, or when a link goes up/down. LSA sequence numbers are used to detect duplicate LSAs, but also serve a second important function in OSPF. With flooding, it is possible for an LSA generated by the source at time t to arrive *after* a newer LSA that was generated by the same source at time $t + \delta$. The sequence numbers used by the source node allow an older LSA to be distinguished from a newer LSA. The age field serves a purpose similar to that of a TTL value. The initial age field value is set to zero and is incremented at each hop as it is flooded, and is also incremented as it sits in a router's memory waiting to be flooded. Although we have only briefly described the LSA flooding algorithm here, we note that designing LSA broadcast protocols can be very tricky business indeed. [RFC 789; Perlman 1999] describe an incident in which incorrectly transmitted LSAs by two malfunctioning routers caused an early version of an LSA flooding algorithm to take down the entire ARPAnet!

4.7.2 Multicast

We've seen in the previous section that with broadcast service, packets are delivered to each and every node in the network. In this section we turn our attention to **multicast** service, in which a multicast packet is delivered to only a *subset* of network nodes. A number of emerging network applications require the delivery of packets from one or more senders to a group of receivers. These applications include